Competitive Adsorption Model for Process Design of Separation and Recovery Method in Porous Type Anion-Exchange Resin

Kousuke Hiromori, Kosei Kanuma and
Naomi Shibasaki-Kitakawa

Department of Chemical Engineering, Tohoku University, 6-6-07 Aoba-yama, Aoba-ku, Sendai-shi, Miyagi 980-8579, Japan

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To optimize operating conditions for separation and recovery method with a porous type anion-exchange resin, a competitive adsorption model was developed by considering mass transfer and ion-exchange reactions of each component. The model well described the concentration profiles adsorbed in the axial direction of the column packed with the resin, which are difficult to determine experimentally, in addition to the concentration profiles in the effluent from the column. The model was also able to predict optimum operating conditions to maximize the recovery yield of the target material and is effective for the process design of the recovery method with the resin.

Introduction

Natural lipid soluble phenolic compounds such as vitamin E are widely used as bioactive compounds with a high antioxidant ability (Tripoli *et al.*, 2005; Peh *et al.*, 2016). Vegetable oils have concentrations of a few percent and are separated and recovered industrially via multi-stage molecular distillation (Bellafiore and Henretta, 2000; Jiang *et al.*, 2006; Posada *et al.*, 2007; Kimura *et al.*, 2009). However, the product is expensive due to large pyrolysis losses under the high temperature conditions. Hence, most of the compounds are synthetic rather than natural (Industry Experts, 2013).

We proposed a novel method for recovering natural vitamin E via adsorption and desorption on an anion-exchange resin. It focused on the difference in acidity of each component (Hiromori *et al.*, 2016, 2018). By relaxing the operating conditions and improving the separation selectivity, the yield and purity of vitamin E were significantly improved. However, the raw material also contains free fatty acid that competitively adsorbs on the resin. If the supply of raw material is too large, vitamin E adsorbed on the resin is eluted out of the column by the free fatty acid that has a higher acidity and larger adsorption energy. Therefore, to recover vitamin E, it is necessary to optimize the supply according to the raw material compositions, and to construct a kinetic model that can quantitatively describe competitive adsorption.

Overshoot has been observed in competitive adsorption, in which an adsorbed component was eluted by other components with a larger adsorption energy and concentrated at a higher value than the raw material (Goodall *et al.*, 2008). Although there are many single-component adsorption models showing a general sigmoidal breakthrough curve, no models for competitive adsorption showing the overshoot curve have been reported. El-Sayed and Chase (2010) constructed a model for the separation of binary proteins by considering the multicomponent Langmuir equation. The model constants were estimated from batch adsorption experiments using a single component system. The constants were then used to simulate flow adsorption in a binary component system. However, the model did not represent the difference in the breakthrough points of each component. This was because the competitive adsorption between the two components was not taken into account. Mota *et al.* (2019) constructed a model based on the bi-Langmuir equation that considers multiple adsorption sites with different adsorption energies. They attempted to predict competitive adsorption in multi-component systems, but could not describe specific overshoot behaviors (Gomes *et al.*, 2018; Mota *et al.*, 2019).

In this research, to optimize the operating conditions for separation and recovery method based on adsorption and desorption in an anion-exchange resin, we constructed a kinetic model that can quantitatively address the competitive adsorption of vitamin E and free fatty acid. Batch competitive adsorption experiments were performed under various operating conditions in the binary system. The model constants were estimated by fitting the model to the experimental results. Next, competitive adsorption and desorption flow experiments were conducted using a column packed with an adsorbent, and the results were compared to the simulations to assess the validity of the model. Finally, to maximize the recovery yield of vitamin E, an optimum

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Correspondence concerning this article should be addressed to K. Hiromori (E-mail address: kousuke.hiromori.e8@tohoku.ac.jp).

amount of raw material for the flow adsorption was predicted by the model.

1. Construction of Competitive Adsorption Model for Flow System

Figure 1 shows a conceptual diagram of competitive adsorption. In this system, the porous type anion-exchange resin, Diaion PA306S (Mitsubishi Chemical Co., Tokyo, Japan), with a low crosslinking density was used as adsorbent (Hiromori *et al.*, 2016). The properties and functional groups of the resin are shown in **Table 1**. The active site is a hydroxyl ion, which exchanges with other ion having a higher acidity. W_{resin} [g-wet] of the resin was packed into a column with inner diameter d [cm] and length L [cm]. A feed solution containing vitamin E (V_EH) and free fatty acid (FaH) was continuously supplied at the bottom of the column at a constant flow rate v_T [cm^3/min]. In the porous structure of the resin, a resin liquid phase was considered in addition to the resin solid and the bulk liquid phases. The following six assumptions are taken into account.

i) The resin particles were spherical and active sites were uniformly distributed.
ii) The external mass transfer resistance of the resin was negligible.
iii) The internal mass transfer resistance of the resin was negligible based on the previous paper (Shibasaki-Kitakawa *et al.*, 2007), reporting that the particle diameter of the resin has little effect on the reaction or adsorption/desorption.
iv) A linear partition equilibrium concentration between the bulk liquid phase C_i and the resin liquid phase $\overline{C_i}$ for each component was considered with partition coefficient H_i.

$$\overline{C_i} = H_i \cdot C_i \tag{1}$$

v) The distribution of each component concentration along the radial direction of the column was negligible.
vi) The dispersion of each component concentration along the column axis direction was considered.

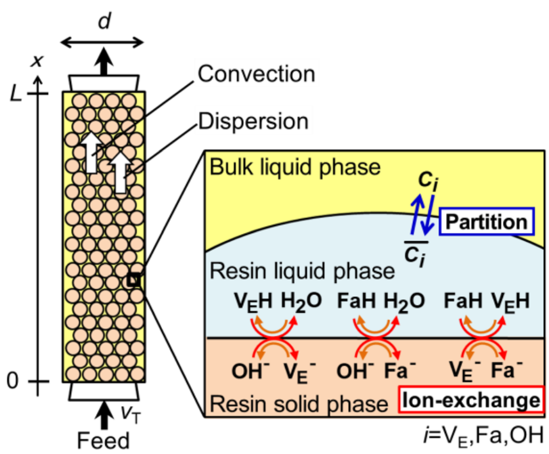

Fig. 1 Conceptual diagram of competitive adsorption

Table 1 Physical properties of ion-exchange resin

Resin	Diaion PA306S
Character	Anion
Matrix type	Porous
Particle diameter	0.23 mm
Ion-exchange capacity	>0.8 [meq/cm^3-resin]
Functionality	Trimethyl amine

Each component in the bulk liquid phase was moved to the resin liquid phase by the partition. Then, the adsorption of V_EH, the adsorption of FaH, and the desorption of V_EH by FaH with a larger adsorption energy occurred via three ion-exchange reactions between the resin liquid and solid phases, as given by Eqs. (2)–(4).

$$\mathrm{V_EH} + \mathrm{S^+(OH^-)} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{S^+(V_E^-)} + \mathrm{H_2O} \tag{2}$$

$$\mathrm{FaH} + \mathrm{S^+(OH^-)} \underset{k_{-2}}{\overset{k_2}{\rightleftarrows}} \mathrm{S^+(Fa^-)} + \mathrm{H_2O} \tag{3}$$

$$\mathrm{FaH} + \mathrm{S^+(V_E^-)} \underset{k_{-3}}{\overset{k_3}{\rightleftarrows}} \mathrm{S^+(Fa^-)} + \mathrm{V_EH} \tag{4}$$

Here, S^+ is a resin skeleton and k_j and k_{-j} are rate constants of forward and reverse reactions, respectively. The equilibrium constants for each ion-exchange reaction $K_{\text{eq}j}$ are defined by the following equations.

$$K_{\text{eq}1} = \frac{k_1}{k_{-1}} \tag{5}$$

$$K_{\text{eq}2} = \frac{k_2}{k_{-2}} \tag{6}$$

$$K_{\text{eq}3} = \frac{k_3}{k_{-3}} \tag{7}$$

By considering consumption and formation in the ion-exchange reactions, the rates of concentration changes in the resin solid phase, q_i, are expressed by Eqs. (8) and (9).

$$\begin{aligned}\frac{dq_{V_E}}{dt} = &-\frac{k_1}{K_{\text{eq}1}} \cdot q_{V_E} \cdot \overline{C_{\text{OH}}} - k_3 \cdot q_{V_E} \cdot \overline{C_{\text{Fa}}} \\ &+ k_1 \cdot q_{\text{OH}} \cdot \overline{C_{V_E}} + \frac{k_3}{K_{\text{eq}3}} \cdot q_{\text{Fa}} \cdot \overline{C_{V_E}}\end{aligned} \tag{8}$$

$$\begin{aligned}\frac{dq_{\text{Fa}}}{dt} = &-\frac{k_2}{K_{\text{eq}2}} \cdot q_{\text{Fa}} \cdot \overline{C_{\text{OH}}} - \frac{k_3}{K_{\text{eq}3}} \cdot q_{\text{Fa}} \cdot \overline{C_{V_E}} \\ &+ k_2 \cdot q_{\text{OH}} \cdot \overline{C_{\text{Fa}}} + k_3 \cdot q_{V_E} \cdot \overline{C_{\text{Fa}}}\end{aligned} \tag{9}$$

Using a total ion-exchange capacity q_{total}, the concentration balance of the active site is given by Eq. (10).

$$q_{\text{total}} = q_{\text{OH}} + q_{V_E} + q_{\text{Fa}} \tag{10}$$

The mass balance equation of component i in the column is expressed by Eq. (11).

$$\varepsilon_b \cdot \frac{\partial C_i}{\partial t} = -u \cdot \varepsilon_b \cdot \frac{\partial C_i}{\partial x} + E_m \cdot \varepsilon_b \frac{\partial^2 C_i}{\partial x^2} - (1-\varepsilon_b) \cdot \varepsilon_p \frac{\partial \overline{C_i}}{\partial t} - (1-\varepsilon_b) \cdot (1-\varepsilon_p) \cdot \frac{\partial q_i}{\partial t} \quad (11)$$

Here, ε_b is the void fraction of the resin fixed bed, ε_p is the porosity of the resin particle, u is the linear liquid velocity, x is an axial coordinate, and E_m is a dispersion coefficient. The left side of Eq. (11) is the rate of change in concentration of the bulk liquid phase. The first and second terms on the right side represent axial mass transfer due to conversion and dispersion in the bulk liquid phase, respectively. The third and fourth terms are the rates of change in concentration of the resin liquid phase $\overline{C_i}$ and the resin solid phase, q_i, respectively. The dispersion coefficient of the fixed bed, E_m, is given by Eq. (12) with the diameter of the resin, R_p, by assuming the particle Peclet number (Fogler, 1999; Levenspiel, 1999).

$$E_m = 2 \cdot u \cdot R_p \quad (12)$$

Before the feed solution is added, there are no components in the bulk liquid and resin liquid phases, and the active sites of the resin solid phase are bound by hydroxide ions. Thus, the initial conditions are given as follows.

$$\text{i.c.}\ ; C_i = 0$$
$$\overline{C_i} = 0$$
$$q_i = \begin{cases} 0 & (i = V_E, \mathrm{Fa}) \\ q_{\mathrm{total}} & (i = \mathrm{OH}) \end{cases} \quad (13)$$

The boundary conditions with the assumption of a closed-vessel are given by Eqs. (14) and (15).

$$\text{b.c. } x = 0;\ C_i = \begin{cases} C_{i,0} & (i = V_E, \mathrm{Fa}) \\ 0 & (i = \mathrm{OH}) \end{cases} \quad (14)$$

$$x = L;\ \frac{\mathrm{d}C_i}{\mathrm{d}t} = 0 \quad (i = V_E, \mathrm{Fa}) \quad (15)$$

There are ten unknown constants: the partition coefficients for each component, H_{VE}, H_{Fa}, H_{OH}, the rate and equilibrium constants for each ion-exchange reaction, k_{1-3} and K_{eq1-3}, and the porosity of the resin particles, ε_p. The constants were estimated by fitting experimental results for batch competitive adsorption, as described below.

2. Estimation of Model Constants in Batch Competitive Adsorption

2.1 Batch competitive adsorption experiments

As a model V_EH, tocopherol in the δ form (approx. 90%, Sigma-Aldrich Corp., U.S.A.) was used. As a model FaH, oleic acid (1st grade, Wako Pure Chemical Industries, Ltd., Japan), which is widely contained in the raw material, was used. Ethanol (1st grade, 99.5%, Wako Pure Chemical Industries, Ltd.) was used as a solvent. The porous type anion-exchange resin, Diaion PA306S, was activated by displacing with hydroxyl ions, as previously described (Hiromori *et al.*, 2016).

In the adsorption experiments, 2 g-wet of the activated resin (OH form) was added to 50 cm^3 of a feed solution in a glass bottle containing a certain concentration of V_EH and FaH. The bottle was sealed and shaken at 150 spm in a water bath at 50°C. The concentrations of V_EH and FaH were varied over the range 0.025–0.075 mol/dm^3, based on the values of the actual raw materials. Samples were taken at specific time intervals, and the V_EH and FaH concentrations were measured. The analysis of each concentration was performed as reported previously (Hiromori *et al.*, 2018).

2.2 Application of the competitive adsorption model

In batch competitive adsorption, there is no liquid flow, that is, zero linear liquid velocity u. Thus, by substituting $u = 0$ into Eq. (11), the following is obtained.

$$\varepsilon_b \cdot \frac{\mathrm{d}C_i}{\mathrm{d}t} = -(1-\varepsilon_b) \cdot \varepsilon_p \frac{\mathrm{d}\overline{C_i}}{\mathrm{d}t} - (1-\varepsilon_b) \cdot (1-\varepsilon_p) \cdot \frac{\mathrm{d}q_i}{\mathrm{d}t} \quad (16)$$

The model constants, H_i, k_j, K_{eqj} and ε_p, in the batch and flow systems were the same. Thus, they were estimated by fitting the model equations, Eqs. (1), (8)–(10) and (16), with three sets of experimental results for batch adsorption. Based on the batch experimental condition, the initial conditions were set as follows.

$$\text{i.c.}\ ; C_i = C_{i0}$$
$$\overline{C_i} = 0$$
$$q_i = \begin{cases} 0 & (i = V_E, \mathrm{Fa}) \\ q_{\mathrm{total}} & (i = \mathrm{OH}) \end{cases} \quad (17)$$

The fitting procedure was similar to that reported previously (Fukumura *et al.*, 2009) and briefly described below. The equation was solved numerically at each time step using the Runge–Kutta method to determine the concentration of component i in each phase. Then, via the Simplex method (Nelder and Mead, 1965), the set of model constants were determined to minimize the summation of the squared absolute errors between the experimental concentration in the bulk liquid phase, $C_{i,\mathrm{exp}}$, and the calculated one, $C_{i,\mathrm{cal}}$.

$$S_{\mathrm{Err}} = \sum_{N_{\mathrm{exp}}} \sum_{N_{\mathrm{time}}} \sum_{i = V_E, \mathrm{Fa}, \mathrm{OH}} (C_{i,\mathrm{exp}} - C_{i,\mathrm{cal}})^2 \quad (18)$$

Here, N_{exp} is the number of experimental conditions and N_{time} is the number of sampling times. The time step of the Runge–Kutta method was 1.0 s.

Figure 2 shows the experimental and fitted results for the bulk concentration of each component under various initial conditions. At all the conditions, the fitted lines were in good agreement with the experimental data. The model adequately showed that the FaH concentration, with the larger adsorption energy, rapidly decreased and became constant. Meanwhile, the concentration of V_EH slightly decreased, and then tended to increase when FaH remained in the bulk liquid phase. This decrease in concentration followed by an increase is a characteristic of competitive adsorption.

The estimated partition coefficients for each component are listed in **Table 2**. The value for water was the largest, which was consistent with previous experiments showing that the resin preferentially takes up highly polar materials (Shibasaki-Kitakawa *et al.*, 2013). The estimated kinetic and equilibrium constants for the three ion-exchange reactions are listed in **Table 3**. The kinetic constants were in the order $k_2 > k_3 > k_1$, and the equilibrium constants were in the same order. Because the ion-exchange reaction was induced by an electrical charge bias, the rate and equilibrium constants were higher for larger differences in acidity between the two components. From the pK_a of each component ($pK_{a,VEH} = 12.6$, $pK_{a,FaH} = 4.8$, $pK_{a,H_2O} = 13.99$) (Spector, 1975; Lide, 2003; Mukai *et al.*, 2007), the differences in acidity for each reaction, ΔpK_a, were calculated and are listed in Table 3. The order was consistent with that of the estimated values. In addition, the estimated porosity of the resin particle was 0.226, which agreed with the nominal value given by the resin manufacturer. Overall, the estimated model constants were reasonable.

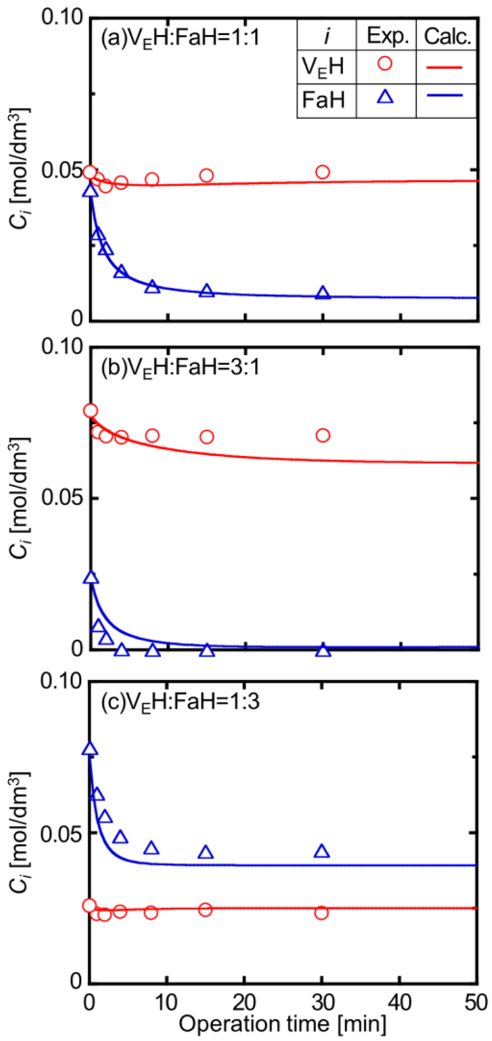

Fig. 2 Experimental and fitted results for batch competitive adsorption under various initial conditions

Table 2 Estimated partition coefficients for each component

Component i	H_i [—]
V_EH	1.96
FaH	2.03
H_2O	5.22

Table 3 Estimated kinetic and equilibrium constants for ion-exchange reactions

Ion-exchange reaction j	Constants		ΔpK_a
	k_j [dm^3/mol/s]	K_{eqj} [—]	
$j=1$ (adsorption of V_EH, Eq. (2))	0.973×10^{-2}	47.4	1.39
$j=2$ (adsorption of FaH, Eq. (3))	11.3×10^{-2}	412	9.19
$j=3$ (desorption of V_EH by FaH, Eq. (4))	4.25×10^{-2}	324	7.80

3. Simulation of Competitive Adsorption and Desorption in Flow System

3.1 Competitive adsorption and desorption flow experiments

An ethanol solution containing V_EH and FaH was used as a model feed solution similar to the above batch adsorption experiments. Each concentration was kept constant at 0.05 mol/dm^3. The 6.7 g-wet resin was packed in a glass column (ILC-FW11-100, inner diameter of 1.1 cm, length of 10 cm, Kiriyama Glass Work Co., Japan), and was then activated by displacing with hydroxyl ions.

Each experiment was performed with adsorption, washing, and desorption steps. In the adsorption step, the feed solution was supplied in an upward flow from the bottom of the column to adsorb the targeted component on the resin. In the washing step, ethanol was supplied in a downward flow from the top of the column to extrude the feed solution remaining in the column. In the desorption step, the desorption solution of 0.43 mol/dm^3 acetic acid in ethanol was supplied in a downward flow from the top of the column to elute the adsorbed components. In each step, the operating temperature was kept constant at 50°C, and the flow rate of feed supply was 1.0 cm^3/min. The direction of the supply solution was determined by considering the density difference between the supply and the remaining solution in the column. The volume of feed solution for the adsorption step was varied over 40–250 cm^3. For the analysis, effluent from the column was collected at specified time intervals, and the concentrations of each component were measured in the same manner as in the batch experiment.

3.2 Simulation of competitive adsorption in a flow system

Competitive adsorption in the flow system was simulated by the model using estimated model constants from the batch system. The simulation conditions were set based on the experimental conditions described above. **Figure 3** shows the calculated and experimental results of the concentration profiles in the effluent during the adsorption step.

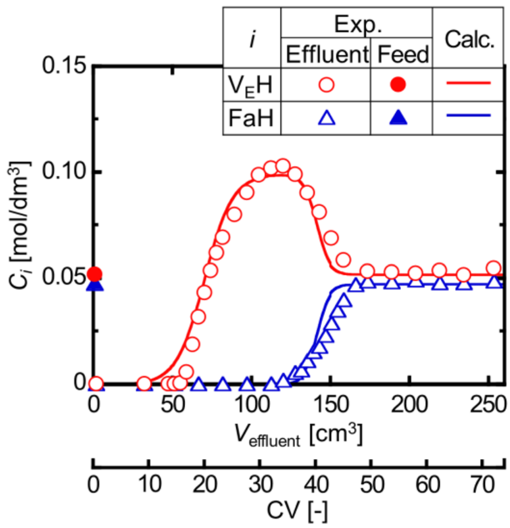

Fig. 3 Experimental and simulated results for concentration profiles in effluent during the adsorption step

The feed supplied volume was 250 cm^3. The abscissa is the total volume of the effluent from the start of the feed supply $V_{effluent}$, which corresponds to the operating time. The co-abscissa denotes the values normalized by the column volume (CV, void volume of the resin bed [cm^3]). For reference, each concentration in the feed solution is indicated by a filled symbol on the ordinate. The simulated lines were in good agreement with the experimental data. In particular, the model adequately simulated overshoot, where V_EH, which had a lower adsorption energy, was up to twice as concentrated as the feed. The model also simulated that after the V_EH concentration reached a maximum, the FaH concentration began to increase and asymptotically approached the feed concentration. This confirmed that the estimated model constants were valid.

The model also could describe the concentration profiles in the bulk liquid, resin liquid, and solid phases along the axial direction of the column, which were difficult to determine experimentally. In particular, the concentration in the resin solid phase indicated the amount adsorbed on the resin, which was likely to be recovered in the desorption step. Therefore, the model could possibly use this concentration to predict the recovery amount.

3.3 Model prediction of optimum adsorption conditions for maximizing vitamin E recovery yield

The concentration profiles along the axial direction of the column were not steady state, and varied with the operating time of the adsorption step. Thus, the profiles at various operating times were simulated under the same condition as in Figure 3. The simulated results for the concentration profiles in the resin solid phase, q_i [mol/dm^3-resin solid] are shown in **Figure 4**. Here, the operation time is presented as the value normalized by CV. The abscissa is the column length. Figure 4(a) shows that FaH was only present near the column inlet and its concentration was zero at $x = 4$ cm. In contrast, V_EH was widely distributed over $x = 2$–10 cm, with

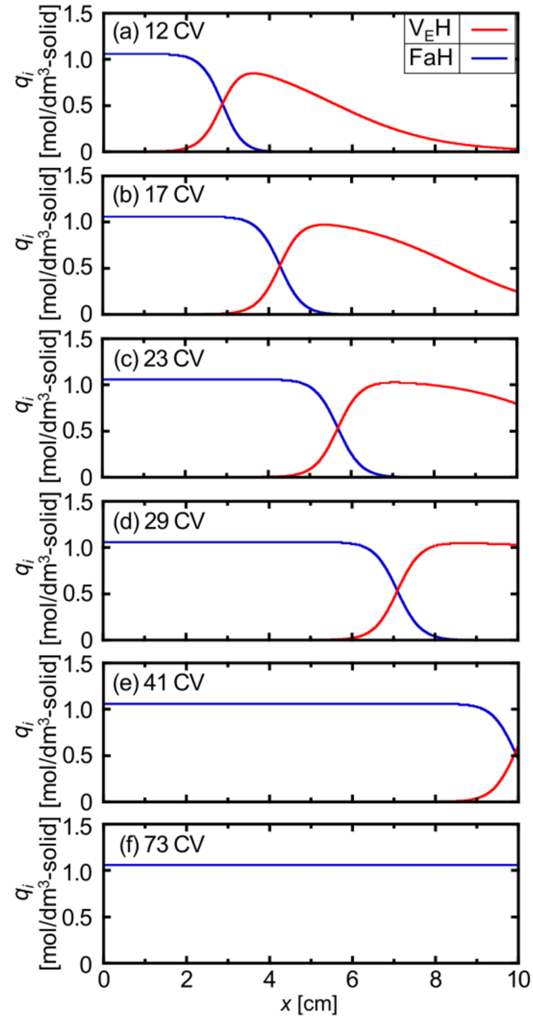

Fig. 4 Simulated results for concentration profiles in the resin solid phase along the axial direction of the column for various supplied volumes

a maximum at 4 cm. When the operating time was increased in Figure 4(b) and (c), the FaH region expanded to $x = 7$ cm, and the V_EH region moved to the outlet as its maximum concentration increased. Further increases in the operating time in Figure 4(d) and (e), enabled the FaH region to reach the outlet and the V_EH flowed out of the column. Finally, as shown in Figure 4(f), only FaH was present.

V_EH and FaH were thus retained on the resin solid phase during the adsorption step, and were recovered during the desorption step. Therefore, the adsorption amount n_i could be calculated by integrating the concentration profile in the resin solid phase q_i, as follows.

$$n_i[\text{mmol}] = \left(\frac{d}{2}\right)^2 \cdot \pi \cdot (1-\varepsilon_b) \cdot (1-\varepsilon_p) \int_0^L q_i \, dx \tag{19}$$

This value would be equal to the actual amount recovered during the desorption step. The model simulations were used to calculate n_i at various feed supplied volumes in the adsorption step. The other simulation conditions were the same as in Figure 3. The calculated values are solid lines in

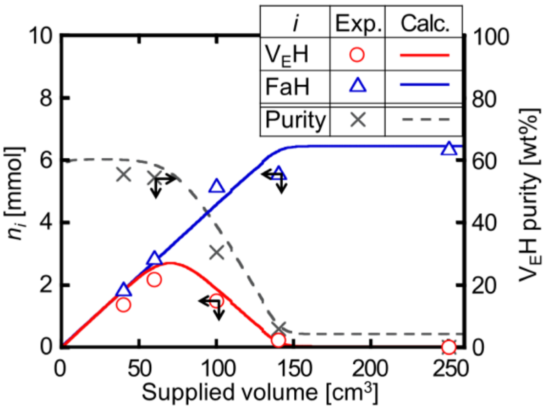

Fig. 5 Experimental and simulated results for recovery amounts during the desorption step

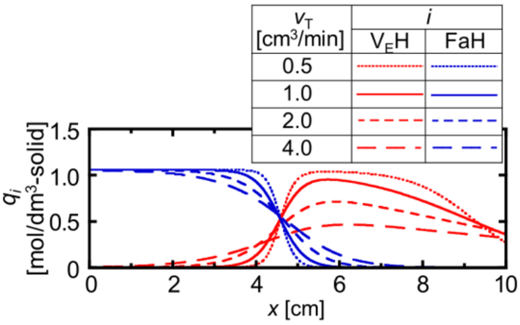

Fig. 6 Effect of feed flow rate on concentration profile in the resin solid phase along the axial direction of the column

Figure 5, compared with the experimental data. The ordinate is the feed supplied volume in the adsorption step. The calculated lines were in good agreement with the experimental data. Thus, the competitive adsorption model could be used to predict the recovery amount in the desorption step for each component. In addition, it was found that the maximum amount of V_EH recovery occurred at a feed supplied volume of 75 cm^3.

Another important indicator was the V_EH purity in the recovered solution during the desorption step. This was calculated by the model based on the amount of each adsorbed component, as follows.

$$V_EH \text{ purity [wt\%]} = \frac{n_{V_EH} \cdot Mw_{V_EH}}{n_{FaH} \cdot Mw_{FaH} + n_{V_EH} \cdot Mw_{V_EH}} \times 100 \tag{20}$$

Herein, Mw_i is molecular weight of component i. The calculated values are shown by a dotted line in Figure 5, while the experimental results are crosses. The V_EH purity was a constant 65 wt% until the maximum V_EH recovery amount was obtained, after which it significantly decreased. Therefore, the V_EH purity was also maximized at the feed supplied volume that maximized the V_EH recovery amount.

Furthermore, the model simulations were used to discuss the effect of the feed flow rate in the adsorption step. The simulated results for the concentrations profiles in the resin solid phase q_i are shown in **Figure 6**. The feed supplied volume was 60 cm^3. As the flow rate increased, the maximum V_EH concentration was lower and the V_EH region became wider. Therefore, the adsorption amount n_i was calculated by Eq. (19) at each flow rate. The calculated values are shown in **Table 4**, along with the Reynolds number Re in the fixed bed (Dwivedi and Upadhyay, 1977) defined by Eq. (21).

$$Re = \frac{\rho \cdot u \cdot R_p}{\mu \cdot \varepsilon_b} \tag{21}$$

Herein, ρ and μ are the density and viscosity of bulk liquid phase, respectively. The values of n_{FaH} were almost constant regardless of the flow rate. On the other hand, n_{VEH} de-

Table 4 Effect of feed flow rate on adsorption amounts of each component

v_T [cm^3/min]	Re [—]	n_i [mmol]	
		V_EH	FaH
0.5	302	2.80	2.92
1.0	604	2.56	2.92
2.0	1209	2.12	2.93
4.0	2418	1.62	2.93

creased with increasing the flow rate. Based on the Re values, the flow condition of the resin bed was considered to be a transition condition ($10 < Re < 2000$) up to 2.0 cm^3/min, and turbulent condition ($Re > 2000$) at 4.0 cm^3/min. Since the model assumes plug flow, the applicable condition was considered to be less than 4.0 cm^3/min.

Hence, the competitive adsorption model enabled predictions of not only concentration profiles in the effluent and along the axial direction of the column during the adsorption step, but also the recovery amount of each component during the desorption step. The model was effective in determining the optimum operating conditions for the adsorption step that maximized the recovery of the target material in the desorption step.

Conclusions

A competitive adsorption model was developed to optimize the operating conditions for the recovery method based on adsorption and desorption on a porous type anion-exchange resin. Considered were mass transfer from the bulk liquid phase to the resin liquid phase and ion-exchange reactions on the resin solid phase. Model constants were estimated from experimental competitive adsorption results in a batch system. Competitive adsorption in a flow system packed with resin was then simulated with the estimated constants. The model adequately simulated the concentration profiles of each component in the effluent during the adsorption step, including overshoot. The model also predicted the concentration profiles of each component adsorbed on the resin along the axial direction of the column at various feed supplied volumes, and predicted the recovery amount for each component during the desorption step.

Thus, the competitive adsorption model was able to predict optimum operating conditions to maximize the recovery yield of the target material, which would be applicable to future process designs of porous type anion-exchange resin.

Acknowledgement

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Nomenclature

C_i	=	concentration of component i in bulk solution phase	[mmol/cm^3]
$\overline{C_i}$	=	concentration of component i in resin liquid phase	[mmol/cm^3]
CV	=	void volume of resin bed	[cm^3]
d	=	inner diameter of column	[cm]
E_{m}	=	dispersion coefficient	[cm^2/min]
H_i	=	partition coefficient of component i	[—]
k_j	=	rate constant of ion-exchange reaction j	[cm^3/mmol/min]
$K_{\mathrm{eq}j}$	=	adsorption equilibrium constant of ion-exchange reaction j	[—]
L	=	column length	[cm]
Mw_i	=	molecular weight of component i	[mg/mmol]
n_i	=	adsorbed amount of component i	[mmol]
N_{exp}	=	number of experimental conditions	[—]
N_{time}	=	number of sampling times	[—]
q_i	=	concentration of component i in resin solid phase	[mmol/cm^3-solid]
q_{total}	=	ion-exchange capacity	[mmol/cm^3-solid]
Re	=	Reynolds number in fixed bed	[—]
R_{p}	=	diameter of ion-exchange resin	[cm]
S_{Err}	=	summation of the squared absolute errors	[mmol^2/cm^6]
t	=	reaction time	[min]
u	=	linear velocity	[cm/min]
V_{effluent}	=	effluent volume from column	[cm^3]
v_{T}	=	flow rate	[cm^3/min]
W_{resin}	=	weight of resin packed into column	[g-wet]
x	=	axial direction in column	[cm]
ε_{b}	=	void fraction in resin bed	[—]
ε_{p}	=	porosity of the resin particle	[—]
μ	=	viscosity of fluid	[g/cm/min]
ρ	=	density of fluid	[g/cm^3]

‹Subscript›

cal	=	calculated value
exp	=	experimental value
i	=	component
j	=	reaction
0	=	initial value or feed

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